Length Baselines and Evidence-Support Triage: An ALCE-to-RAGTruth Audit

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Abstract

Exploratory comparisons on an exposed RAGTruth set depend on the estimand: answer length ranks higher on pooled average precision (AP), while NLI adds conditional incremental signal when combined with length and generator. On 900 QA answers from 150 question groups, raw length minus coverage NLI has AP difference 0.1045 (conditional 95% interval [0.0255, 0.1848]) and AUC difference 0.0279 [-0.0291, 0.0831]. Equal-weight generator macro AUC instead favors NLI but is unstable because GPT-4 has one positive; removing GPT-4 yields an interval crossing zero. Five-fold question-group cross-fitting gives AP 0.3948 for length plus generator and 0.4245 after adding NLI, with paired gain 0.0297 [0.0109, 0.0486]. These intervals resample fixed predictions and omit training and fold-selection uncertainty. All contrasts use 5,000 paired question-group draws under a committed post-exposure plan; none are confirmatory or multiplicity-adjusted. A character-span-to-token check yields heterogeneous generator-specific rates and a sparse-stratum boundary problem, without settling causal exposure. Earlier ALCE source calibration remains context, not a pooled endpoint. Restricted texts remain local; derived predictions, all planned contrasts, failures and execution limitations are released. The incremental signal is a hypothesis for fresh confirmation, not a general detector advantage or deployment guarantee.

1. Research question and bounded contribution

The primary question in this revised manuscript is whether the tested evidence-reading rankings establish an advantage beyond a trivial answer-length control on this particular benchmark. The updated evidence depends on the comparison: raw length has higher pooled AP under conditional exploratory intervals, but a fixed cross-fitted model gains AP and AUC when NLI is added to length and generator. Coverage NLI has higher raw AUC in four of six generator strata. The incremental result is a hypothesis for fresh confirmation; neither this audit nor its conditional intervals establish a general method ranking, reliable deployment or statistical equivalence.

The original study asked whether source-calibrated policies retain recall and manageable workload across ALCE sentences and RAGTruth whole answers, with a diagnostic comparison of bounded and coverage NLI representations. That protocol preceded target scoring. Length and generator controls were introduced after the outcomes and critiques were exposed, and motivated the revised framing. They are post-hoc observational diagnostics; the original source-transfer question remains secondary context. Stronger matched detectors and a fresh frozen confirmation set remain untested.

ALCE evaluates answers with citations and provides human sentence-support annotations (Gao et al., 2023). RAGTruth provides manually annotated hallucination spans in generated responses grounded in supplied material (Niu et al., 2024). Their annotations serve related purposes but are not identical constructs. This manuscript reports an empirical audit; it does

not introduce a new learning algorithm, claim a first use of window aggregation, establish state-of-the-art performance, or establish a comprehensive novelty result. Previously exposed ALCE comparisons are historical context. The new RAGTruth protocol was committed before its model outcomes were computed.

2. Data, labels and independence

The ALCE annotation file contains 2,604 sentences from 200 supplied question groups across ASQA and ELI5. All available model responses to the same supplied question stay together. Source train and validation sets follow the existing seed42 split: ASQA has 639 training sentences from 81 questions and 172 validation sentences from 19 questions; ELI5 has 1,430 training sentences from 81 questions and 363 validation sentences from 19 questions. Unsupported is the positive class, defined as one minus the binary human sentence-support label. The other domain is an exposed historical target, never used to calibrate the new external experiment.

The independent external selection uses the published RAGTruth test split and task_type QA only. Every eligible response is retained, without filtering by generator, quality, label or score. A complete question group would be excluded if its lowercased whitespace-normalized question exactly matched any ALCE question. There were zero such exclusions, leaving 900 answers and 150 source_id groups. A positive answer has at least one supplied hallucination span; zero spans is negative under this released annotation rule, not a certificate of truth. All supplied quality flags, including imperfect outputs, remain available only as descriptive strata. Neither question text, model identity, quality flags, prompts nor human annotation spans enter model inputs. Only the response and its supplied evidence do.

The ALCE file SHA-256 is cfed9293752413d7c7631f36524dd4ee9ef58b209cdf9c63f6fc1e280b43cca6. RAGTruth is pinned to commit c103204b9ce28d6bbad859304bf30de72b8ed8fe; the complete input hashes are stored in the protocol and measured result. The annotation repository has an MIT license, while its underlying MS MARCO web material has separate noncommercial research terms. Raw questions, passages, answers and hallucination spans are not redistributed. Exact question matching cannot establish semantic independence. Unknown model-pretraining overlap, corpus selection and annotation errors remain unresolved.

3. Frozen policies and evaluation

The lexical policy uses evidence-token overlap, number of citations, unique claim-token count, unique evidence-token count, unmatched-number fraction and a missing-citation indicator. A train-only standard scaler and balanced logistic regression fit each ALCE source separately. The highest validation threshold retaining at least 90% source unsupported recall is fixed before external scoring. The external complete answer is represented with one supplied evidence citation. This changes feature distributions; it is not a sentence-to-answer invariant. The implementation requires the previously measured lexical ALCE target AP and source threshold to reproduce before external inference.

The NLI comparator is cross-encoder/nli-deberta-v3-small at immutable revision fa2804872c3b4bd748f38c0185cc85775361e735, with hash-checked ONNX/tokenizer files and an Apache-2.0 model card. The model card identifies SNLI/MultiNLI training and label order contradiction, entailment, neutral. Unsupported risk is one minus the entailment softmax score. It is a model score, not a calibrated real-world probability. The bounded representation allows

512 total pair tokens and caps the answer at 120 tokens, truncating evidence first. Earlier ALCE-calibrated thresholds are reused without external adaptation.

The coverage diagnostic uses these identical weights. Evidence windows have 384 tokens and stride 320; answer windows have 120 tokens and stride 100. An end-aligned final window ensures that every original token appears in a window. Each answer window receives its maximum entailment score across all evidence windows. The whole-answer risk is the maximum unsupported score across answer windows. The policy uses 1,836 inference pairs for 900 external answers, with zero re-encoding truncations. Source validation separately calibrates this new representation, using the same 90% recall rule. Window dimensions and aggregation were frozen before target scoring. This method can split joint semantics and is intrinsically length-dependent; it is a diagnostic adaptation, not a claimed novel detector.

We report average precision (AP), ROC AUC, recall, supported-answer false-positive rate (FPR), review fraction and missed positives. AP is scikit-learn average precision, not trapezoidal PR area. Thresholds use source validation only. Review-all and review-none controls expose the workload extremes. Uncertainty uses 300 complete-source_id bootstrap resamples, seed 20261019, with paired AP differences on identical rows. These percentile intervals condition on fitted policies and the sampled corpus. They do not include model-fitting, annotation, source-selection or pretraining uncertainty. Multiple descriptive contrasts are not a family-wise corrected significance study.

4. Measured results

Table 1 gives the historical ALCE sentence-transfer results. These rows are not pooled with whole-answer RAGTruth metrics.

Direction	Policy	AP	Recall	Review	Missed
ASQA to ELI5	Lexical	0.919	92.84%	70.33%	64
ASQA to ELI5	Bounded NLI	0.838	81.54%	63.97%	165
ELI5 to ASQA	Lexical	0.857	86.33%	52.53%	38
ELI5 to ASQA	Bounded NLI	0.775	95.32%	84.96%	13

Table 2 gives the new whole-answer RAGTruth results. Source denotes the calibration policy, not a target subgroup. Bounded and coverage NLI risk rankings are the same across source rows; only their source-calibrated thresholds change. The target positive prevalence is 17.8%.

Source	Policy	AP	ROC AUC	Recall	FPR	Review	Missed
ASQA	learned_lexical	0.281	0.709	100.00%	98.65%	98.89%	0
ASQA	bounded_nli	0.210	0.602	33.12%	27.03%	28.11%	107
ASQA	coverage_nli	0.281	0.699	60.00%	31.08%	36.22%	64
ELI5	learned_lexical	0.218	0.620	81.88%	64.05%	67.22%	29
ELI5	bounded_nli	0.210	0.602	79.38%	64.05%	66.78%	33
ELI5	coverage_nli	0.281	0.699	91.25%	66.49%	70.89%	14

The bounded representation truncated evidence for 151 of 900 answers and truncated claims for 458. Consequently, a positive answer annotation can refer to text not observed by this policy. Covering all tokens improves AP from 0.210 to 0.281 for these fixed weights, but does not isolate token coverage as a causal explanation: aggregation and representation change together. With ELI5 calibration, coverage misses 14 positives rather than bounded NLI's 33, while increasing the review fraction from 66.8% to 70.9% and supported-answer FPR from 64.1% to 66.5%. With ASQA calibration, coverage misses 64 positives; its recall is 60.0%. There is no transferred 90% recall guarantee. The ELI5 coverage recall is exactly 146/160 = 91.25%; table percentages use two decimals to avoid inconsistent one-decimal rounding.

Table 3 gives the conditional 95% percentile intervals for AP and its paired difference from the corresponding lexical source policy. These are descriptive intervals from 150 question clusters, not 900 independent answer draws.

Source / Policy	AP interval	AP difference from lexical
ASQA / learned_lexical	[0.231, 0.344]	reference
ASQA / bounded_nli	[0.174, 0.255]	[-0.121, -0.025]
ASQA / coverage_nli	[0.227, 0.337]	[-0.053, 0.054]
ELI5 / learned_lexical	[0.181, 0.264]	reference
ELI5 / bounded_nli	[0.174, 0.255]	[-0.046, 0.032]
ELI5 / coverage_nli	[0.227, 0.337]	[0.017, 0.117]

The coverage-versus-ELI5 lexical AP interval excludes zero under this conditional procedure. The coverage-versus-ASQA lexical difference includes zero. Review-all obtains recall 100%, FPR 100% and review 100%; review-none obtains recall 0%, FPR 0% and review 0%. Both constant-score controls have AP equal to prevalence, 0.178. Neither is a practical efficiency success. The ASQA lexical policy's perfect recall with 98.9% review approaches the review-all extreme, so recall alone obscures its operational weakness.

4.1. Post-hoc generator and length controls

These controls were added after the target outcomes and the owner's critique were visible. Every generator is retained, including GPT-4 with only one positive. Table 4 reports within-generator length and coverage NLI ROC AUC; AP and both length directions are retained in review-diagnostics.json. Sparse positive strata are descriptive only.

Generator	Answers	Positives	Length AUC	Coverage NLI AUC
gpt-3.5-turbo-0613	150	5	0.664	0.818
gpt-4-0613	150	1	0.074	0.638
llama-2-13b-chat	150	36	0.684	0.601
llama-2-70b-chat	150	35	0.570	0.633
llama-2-7b-chat	150	52	0.600	0.638
mistral-7B-instruct	150	31	0.773	0.655

Average ranks handle ties within each generator and are divided by that generator's answer count, then pooled over all 900 rows. This uses the unlabeled target score distribution and generator identities: a transductive diagnostic, not a source-frozen deployment policy. The identical transformation is applied to coverage NLI for symmetry. Table 5 includes both transformed and untransformed scores; a normalized length score versus raw NLI is not a matched normalization comparison.

Ranking	AP	ROC AUC
Raw length, longer risk	0.386	0.727
Raw length, shorter risk	0.117	0.273
Raw coverage NLI	0.281	0.699
Within-generator length ranks	0.296	0.632
Within-generator coverage ranks	0.236	0.625

Length has ROC AUC above 0.5 in five generator samples. GPT-4 instead has length AUC 0.074 and only one positive; this unstable, opposite-direction result is retained rather than excluded. Normalizing within generators changes cross-generator comparisons and pooled AP/AUC; the reduction from raw length AUC 0.727 to rank-normalized AUC 0.632 is not an additive estimate of a generator effect. Section 4.2 reports newly computed conditional intervals under the committed exploratory plan; no equivalence margin or confirmatory test was specified. Longer answers may also have more opportunities for at least one annotated hallucination, without greater per-claim error. These data do not separate that exposure effect from semantic quality or generator behavior.

4.2. Exploratory paired contrasts and incremental value

All results in this section use the exposed target. The original plan was committed in 9002e048b42cec344165f8f8e4fe94f59a0e8bd6 before these additional calculations; its text remains unchanged. Exactly 5,000 complete-question bootstrap draws, seed 20261025, are shared across scores and populations. Intervals are conditional 95% percentile summaries, uncorrected for multiple contrasts. AP means scikit-learn average precision. Positive length-minus-NLI differences favor length; positive M1-minus-M0 differences favor the augmented model. Every planned contrast, including zero-crossing intervals, is reported.

Table 6. Exploratory exposed-target paired differences. Rank transformations are recomputed within every sampled generator, retaining duplicate-group multiplicity. Macro averages weight generators equally and require every included stratum to contain both classes. Counts show valid/invalid draws; invalid draws are never replaced.

Population	Contrast	AP diff [95%]	AUC diff [95%]	Valid / invalid
All six	Raw length - NLI	0.1045 [0.0255, 0.1848]	0.0279 [-0.0291, 0.0831]	5000 / 0
All six	Ranked length - NLI	0.0594 [0.0005, 0.1214]	0.0068 [-0.0549, 0.0709]	5000 / 0
All six	Macro length - NLI	0.0866 [-0.0083, 0.1786]	-0.1029 [-0.1730, -0.0326]	3156 / 1844
All six	M1 - M0	0.0297 [0.0109, 0.0486]	0.0170 [0.0036, 0.0307]	5000 / 0
No GPT-4	Raw length - NLI	0.1043 [0.0256, 0.1827]	0.0486 [-0.0107, 0.1061]	5000 / 0
No GPT-4	Ranked length - NLI	0.0734 [0.0018, 0.1466]	0.0117 [-0.0525, 0.0772]	5000 / 0
No GPT-4	Macro length - NLI	0.1061 [-0.0059, 0.2160]	-0.0107 [-0.0925, 0.0687]	4972 / 28
No GPT-4	M1 - M0	0.0285 [0.0105, 0.0470]	0.0211 [0.0045, 0.0380]	5000 / 0

Raw length has a positive pooled AP difference in this conditional exploratory analysis, whereas both raw and rank-normalized AUC intervals cross zero. The equal-weight macro AUC favors NLI with all generators but crosses zero after GPT-4 is removed. The complete-sample macro interval is conditional on 3,156 valid draws and is particularly unstable because GPT-4 has one positive. These are different estimands, not a basis for selecting a favorable ranking or establishing general superiority or equivalence.

Table 7. Exploratory within-generator length-minus-NLI contrasts. The five retained per-generator contrasts are numerically identical in the GPT-4-excluded sensitivity because their rows and shared question draws are unchanged. Sparse-stratum intervals condition on valid draws; the one-positive GPT-4 interval is not reliable population inference.

Generator	AP diff [95%]	AUC diff [95%]	Valid	Invalid
gpt-3.5-turbo-0613	0.1515 [-0.2636, 0.6067]	-0.1538 [-0.4257, 0.1164]	4972	28
gpt-4-0613	-0.0110 [-0.0414, -0.0080]	-0.5638 [-0.6545, -0.4698]	3174	1826
llama-2-13b-chat	0.1029 [-0.0430, 0.2605]	0.0837 [-0.0494, 0.2152]	5000	0
llama-2-70b-chat	-0.0157 [-0.1665, 0.1346]	-0.0622 [-0.1950, 0.0707]	5000	0
llama-2-7b-chat	0.0159 [-0.1250, 0.1608]	-0.0384 [-0.1630, 0.0902]	5000	0
mistral-7B-instruct	0.2759 [0.0931, 0.4267]	0.1172 [-0.0144, 0.2427]	5000	0

M0 is L2 logistic regression using log1p(answer tokens) and generator indicators; M1 adds continuous coverage-NLI risk. Five fixed question-group folds keep preprocessing and fitting inside training groups. Neither model is a source-only policy: both train on other exposed-target groups. Out-of-fold probabilities are pooled for scoring. The paired bootstrap treats these predictions as fixed and omits refitting, fold-selection and training uncertainty; overlapping fold training sets further limit interpretation.

Table 8. Exploratory pooled out-of-fold model scores. The paired gains and all intervals appear in Table 6.

Population	M0 AP	M1 AP	M0 AUC	M1 AUC
All six	0.3948	0.4245	0.7747	0.7917
No GPT-4	0.3981	0.4265	0.7238	0.7449

NLI contributes a positive conditional incremental signal under these fixed specifications, including after GPT-4 removal. That observation prevents a blanket conclusion that evidence reading adds nothing beyond length and generator. It is a hypothesis for genuinely fresh confirmation, not an established general finding or a source-transfer improvement. No configuration was tuned after these outcomes.

4.3. Bounded annotation-token exposure check

The pinned RAGTruth response file and coverage tokenizer were hash-verified. All 235 retained annotation spans exactly matched Unicode character slices, and all 900 full-answer token lengths matched cached counts. Overlapping spans were merged; a non-special answer token counted once if its offsets overlapped any annotated character span. No answers had zero tokens. This is annotated token coverage, not per-claim error or a certificate of semantic correctness.

Table 9. Exploratory generator-specific rates and length-adjusted binomial-logit slopes. Total rate is annotated tokens divided by all answer tokens; mean rate averages answer fractions. The interaction model uses a separate intercept and log1p(length) slope per generator. Predicted fractions are at that generator's observed length quartiles. Full predicted-fraction intervals, token totals, length distributions, row counts and every saved fit are released in token-rate-analysis.json and token-rate-resamples.json.gz.

Generator	Total rate	Mean rate	Length slope [95%]	Predicted Q1 / Q2 / Q3	Accepted draws
gpt-3.5-turbo-0613	1.211%	0.703%	3.7392 [-1.0407, 13.7289]	0.024% / 0.187% / 0.658%	5000
gpt-4-0613	0.111%	0.204%	-2.3550 [-3.9183, 0.0000]	0.136% / 0.078% / 0.050%	5000
llama-2-13b-chat	6.332%	5.529%	0.9522 [-0.0737, 2.1350]	4.001% / 5.310% / 6.859%	5000
llama-2-70b-chat	7.317%	5.726%	2.1824 [-0.1413, 4.2142]	2.356% / 4.262% / 6.894%	5000
llama-2-7b-chat	7.486%	6.987%	0.6803 [-0.3080, 1.6951]	5.895% / 6.875% / 8.027%	5000
mistral-7B-instruct	6.845%	4.736%	1.8760 [0.2603, 3.5291]	2.002% / 3.826% / 6.242%	5000

All 5,000 token-model fits met the fixed numerical acceptance criteria, with no reported fit failures; four exponential-overflow runtime warnings were retained in the execution notes. Numerical acceptance does not establish identification. In 1,826 draws GPT-4 had no annotated token; its stratum parameters approach a boundary despite finite convergence. The analogous GPT-3.5 count was 28. GPT-4 slope intervals are therefore unreliable numerical summaries, and

GPT-3.5 is also sparse. No retrospective boundary exclusion or replacement fit was performed. The dated execution notes document this limitation separately from the unchanged plan.

The point slopes are positive for five generators and negative for GPT-4, with broad intervals. These differences do not prove or refute the exposure explanation. Span boundaries and annotation conventions can differ across generators; conditioning on answer length does not remove those differences or unmeasured confounding. Question-group refitting supplies descriptive uncertainty, not independent-token standard errors or causal inference.

The token implementation details were fixed in a separate dated amendment before token outcomes; the primary analysis plan remains intact. Runtime notes and boundary diagnostics were added after the run and are labeled as such. Passing tests and numerical convergence check implementation behavior; they do not validate statistical conclusions.

Contrast inventory: original source-only threshold transfer and the frozen coverage representation belong to the earlier external protocol; historical 300-draw comparisons remain unchanged. Raw, rank-normalized and macro length contrasts, within-generator intervals, the GPT-4-excluded sensitivity, cross-fitted incremental value and annotated-token rates are all exposed-target exploratory extensions. No confirmatory contrast or equivalence margin has been tested.

5. Interpretation and alternative explanations

The revised title focuses on baseline adequacy in this ALCE-to-RAGTruth audit. Pooled AP favors raw length in the conditional exploratory comparison, while the fixed target-supervised incremental model gains signal from NLI. This mixed result does not support either a general length-superiority claim or a claim that NLI adds nothing. Failure of source recall and review workload to transfer is supporting context, not a universal transfer theorem.

The within-generator controls in Section 4.1 weaken a semantic-support interpretation without identifying a causal explanation. The five-fold complete-question cross-fitted generator prior from the earlier response has AP 0.274, ROC AUC 0.685. It uses target labels from other question groups; unlike the label-free rank transform, it is a target-supervised diagnostic and not a source-only comparator.

Coverage thresholds are 0.989811 for ASQA (172 sentences, 19 questions) and 0.451716 for ELI5 (363 sentences, 19 questions). They are empirical source recall cutoffs, not calibrated probabilities. Different source score distributions can produce different cutoffs, but the cached data do not establish why this gap occurs. Calibration sensitivity across independent question-group splits has not been measured; a 90% empirical source target is not a population recall guarantee. High historical sentence-level AP does not transfer unchanged to this answer-level benchmark. The target differs in generator mixture, response length, citation presentation, evidence and positive-label definition. These shifts occur together. Accordingly, the numerical AP drop cannot be attributed solely to domain shift, answer granularity or model architecture. A controlled unit-matched annotation study would be needed to separate those explanations.

AttrScore directly studies attribution errors and evaluates prompted and fine-tuned models (Yue et al., 2023). AttributionBench consolidates attribution evaluation and documents persistent failure modes (Li et al., 2024). These are close prior work, not systems reproduced here. Their task-specific F1 scores cannot be compared numerically with this audit's AP. REVIEW_RESPONSE_PROTOCOL.md records the required label mapping, checkpoint/license

verification, context policy and source-only calibration for a matched AttrScore experiment, which remains unexecuted.

The small NLI model is an accessible open comparator, not a comprehensive current-model tournament. Target fine-tuning, prompt-based commercial judges, larger task-specific detectors and learned evidence retrieval are not tested. The source-only design answers a transfer question; it does not estimate the best achievable supervised RAGTruth performance. Comparison with the original RAGTruth paper's tuned systems would require their matched splits, units and procedures. No such numerical ranking is claimed.

Window aggregation has opposing failure modes. A maximum across evidence may find a spurious supporting fragment, while a maximum risk across answer windows may penalize long otherwise supported answers. Splitting text can destroy conjunctions or reference resolution. No calibrated probability, reliable factual verdict or educational learning benefit follows from these scores. Generator and quality strata are released descriptively, without outcome-selected exclusions or selecting a preferred policy after inspecting a favorable stratum.

6. Reproducibility, ethics and limitations

Protocols, source hashes, original code, integrity tests, individual derived risks and aggregate outputs accompany the manuscript in Sunnyvulli/Data-Engineer_AI/research-suite. The external protocol is fixed at commit cf1c7b96b7aee262fd64816377dbcf04168ba5e7. Run python -m research_suite.external_support with the pinned local data and model directory after installing documented dependencies. Hash-validated caches permit interrupted inference to resume. The recorded elapsed time is a resumed invocation, not a reconstructed end-to-end compute bill; no total from discarded process time is fabricated. CPU inference uses four intra-operation threads and batch 4; no paid API or GPU was used.

Integrity checks cover hidden-label exclusion, complete token-window coverage, grouped paired resampling and existing causal/split controls. Passing tests cannot validate statistical conclusions or prove absence of defects. Conditional fixed-prediction intervals omit training and fold-selection uncertainty, and the sparse GPT-4 token-rate fits have boundary nonidentification despite numerical convergence. The four-perspective review is a self-audit generated within this project, not independent expert peer review. No participant study, production deployment, manually adjudicated new labels or acceptance by a journal is represented.

The named human author must inspect the artifacts, verify the claims and confirm substantive authorship and the final text before external submission. AI is not listed as an author. No institutional affiliation, funding award or external reviewer is invented.

7. Conclusion

On this exposed RAGTruth whole-answer set, pooled AP conditionally favors raw length over coverage NLI, but generator-stratified comparisons are mixed and adding NLI to a fixed cross-fitted length-plus-generator model yields a positive conditional incremental signal. That signal is a hypothesis for fresh confirmation, not a general method advantage. Annotated-token rates do not settle a causal exposure explanation, and sparse-generator boundary fits limit their interpretation. These exposed-target analyses establish neither equivalence, causal decomposition nor deployment validity. The original source-threshold workload failures remain relevant, but stronger matched comparators, calibration/window studies, fresh confirmation and independent human review are still required.

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